

EXPERIMENT 11

Title

Reading CSV File and Performing RDD Map Operation using Apache Spark

Aim

To read a CSV file, create RDD, apply map transformation, and display the output using Apache Spark.

Software Requirements

- Apache Spark
- Hadoop
- Java
- CSV File

RDD (Resilient Distributed Dataset) is the basic data structure in Apache Spark. Spark reads data from CSV files and performs transformations such as map operation to process data efficiently.

Procedure

Step 1: Create CSV File

Create a file named:

```
student.csv
```

Add the following data:

```
1,Arun,80
2,Bala,70
3,Chitra,90
4,David,60
```

Save the file.

Step 2: Open Spark Shell

```
spark-shell
```

Step 3: Load CSV File as RDD

```
val data = sc.textFile("student.csv")
```

Step 4: Apply Map Function

```
val result = data.map(x => x.toUpperCase())
```

Step 5: Display Output

```
result.collect().foreach(println)
```

Program

```
val data = sc.textFile("student.csv")

val result = data.map(x => x.toUpperCase())

result.collect().foreach(println)
```

Output

```
1,ARUN,80
2,BALA,70
3,CHITRA,90
4,DAVID,60
```

Result

Thus the CSV file was successfully loaded as RDD and map operation was performed using Apache Spark.

EXPERIMENT NO: 12

Spark CSV Filter Operation using Apache Spark

Aim: To read a CSV file and perform filter operation using Apache Spark.

Software Requirements

- Apache Spark
- Hadoop
- Java
- CSV File

Filter transformation in Spark is used to select only the required records from the dataset based on a condition.

Spark reads the CSV file as RDD and filters the data efficiently.

Procedure

Step 1: Create CSV File

Create file:

```
student.csv
```

Add data:

```
1,Arun,80  
2,Bala,70  
3,Chitra,90  
4,David,60
```

Save in:

```
C:\spark\data\student.csv
```

Step 2: Open Spark Shell

```
spark-shell
```

Step 3: Read CSV File

```
val data = sc.textFile("C:/spark/data/student.csv")
```

Step 4: Apply Filter Operation

```
val result = data.filter(x => x.contains("80"))
```

Step 5: Display Output

```
result.collect().foreach(println)
```

Program

```
val data = sc.textFile("C:/spark/data/student.csv")  
val result = data.filter(x => x.contains("80"))  
result.collect().foreach(println)
```

Output

```
1,Arun,80
```

Result

Thus the CSV file was successfully filtered using Apache Spark.

